# Lesson 5.8 - Compound Probability of Independent Events

> **Video:** Compound Probability of Independent Events - Coins & 52 Playing Cards by The Organic Chemistry Tutor **Chapter:** 5 - Introduction to Probability

## Rule

For **independent** events, multiply:

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

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## Example 1 - coins

**(a) Two heads in two flips**

$$P(HH) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

Check with the sample space $\{HH, HT, TH, TT\}$: 1 of 4 outcomes ✓

**(b) Heads, tails, heads, in that order, in three flips**

$$P(HTH) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Only 1 of the 8 outcomes ($HHH, HHT, HTH, HTT, THH, THT, TTH, TTT$) matches.

**(c) Two tails and one head, in any order** The matching outcomes are $HTT$, $THT$, $TTH$:

$$P = \frac{3}{8}$$

> **Does order matter?** In (b) a specific sequence is required, so only one outcome counts. In (c) any arrangement counts, so add up all the matching outcomes.

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## A standard 52-card deck

| Suit | Color | Cards |
|---|---|---|
| ♦ Diamonds | Red | A, 2-10, J, Q, K (13) |
| ♥ Hearts | Red | 13 |
| ♠ Spades | Black | 13 |
| ♣ Clubs | Black | 13 |

- 4 suits × 13 = **52** cards; **26 red**, **26 black**; no jokers.
- Each rank (e.g. queen) appears **4** times, once per suit.

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## Example 2 - cards

**(a) P(queen)**

$$\frac{4}{52} = \frac{1}{13} \approx 0.0769 = 7.69\%$$

**(b) P(black card numbered greater than 3 and at most 9)** The numbers are 4, 5, 6, 7, 8, 9 (6 values), and there are two black suits (♠ and ♣):

$$\frac{6 \times 2}{52} = \frac{12}{52} = \frac{3}{13}$$

**(c) P(red king first, then a diamond), *without* replacement**

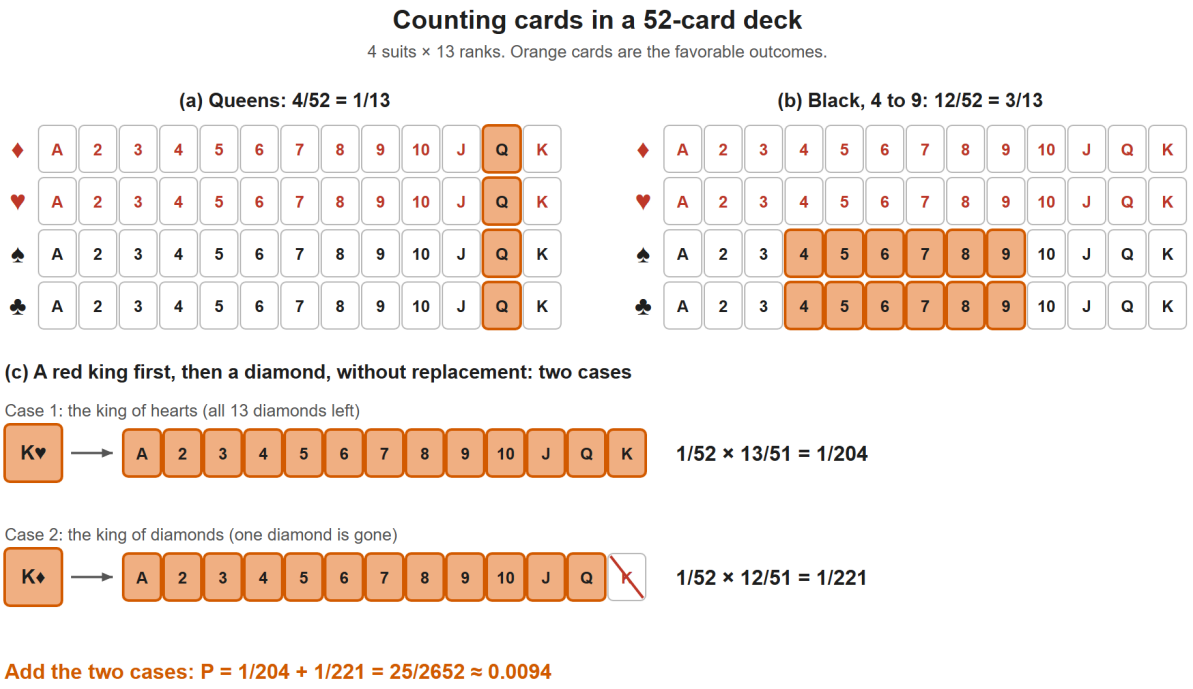
This is **dependent**: how many diamonds are left depends on *which* red king was drawn. Split it into two cases and add them (the cases are mutually exclusive):

| Case | First card | Second card | Probability |
|---|---|---|---|
| 1 | King of ♥ ($\frac{1}{52}$) | any ♦; all 13 are still there ($\frac{13}{51}$) | $\frac{1}{52} \cdot \frac{13}{51} = \frac{1}{204}$ |
| 2 | King of ♦ ($\frac{1}{52}$) | any ♦; only 12 are left ($\frac{12}{51}$) | $\frac{1}{52} \cdot \frac{12}{51} = \frac{1}{221}$ |

$$P = \frac{1}{204} + \frac{1}{221} = \frac{13}{2652} + \frac{12}{2652} = \frac{25}{2652} \approx 0.0094$$

(Common denominator: $2652 = 4 \cdot 3 \cdot 13 \cdot 17$.)

> Writing $P(\text{red king}) \times P(\text{diamond}) = \frac{2}{52} \times \frac{13}{51}$ would be **wrong**, because the second probability depends on which king was taken.




Counting favorable cards in the deck. For part (c) the number of diamonds left depends on which red king was taken first, so the two cases are worked out separately and added.

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## Key takeaways

- Independent events: multiply the probabilities. $n$ fair coin flips in a specific order: $(1/2)^n$.
- "In any order" → count all the arrangements that work.
- Without replacement → dependent. If the second probability depends on *which* first outcome happened, split into cases and add.